No. of Functional Features Included in the Solution

Date	24 July 2026
Team ID	SWUID2026211863
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Module	Accepts soil and environmental parameters such as N, P, K, temperature, humidity, pH, and rainfall from the user.	Frontend (HTML/CSS/Bootstrap)	Done	10%
2	Data Preprocessing	Cleans the dataset, handles missing values and outliers, and prepares data for model training.	Data Processing	Done	15%
3	Exploratory Data Analysis	Performs univariate, bivariate, and multivariate analysis to understand the dataset.	Data Analysis	Done	10%
4	Machine Learning Model Training	Trains and evaluates multiple machine learning algorithms to identify the best-performing model.	ML Module	Done	20%

8	Testing & Validation	Evaluates model performance and verifies application functionality before deployment.	Testing Module	Done	5%
5	Crop Recommendation Prediction	Predicts the most suitable crop using the trained machine learning model based on user inputs.	Prediction Engine	Done	20%
6	Flask Web Application	Connects the frontend with the backend and displays prediction results in real time.	Backend (Flask)	Done	15%
7	Model Persistence	Saves the trained machine learning model using Pickle (.pkl) for future predictions.	Model Storage	Done	5%

Feature Summary:

Metric	**Count / Value**
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	**Category**	**Features in Category**	**Example Features**
1	User Interface (UI)	User Input Forms, Result Display	HTML, CSS, Bootstrap

2	Backend / Logic	Data Preprocessing, ML Prediction, Flask Backend	Python, Flask, Scikit-learn
3	Database / Storage	Crop Dataset, Trained Model	CSV Dataset, Pickle (.pkl)
4	API / Integration	Flask Routing and Prediction Integration	Flask API
5	Security / Authentication	Input Validation and Error Handling	Input Validation, Exception Handling